LE NGUYEN GIA HUNG

AI Research Engineer · Time-Series & Multimodal XAI

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EDUCATION

FPT University Ho Chi Minh City, Vietnam
Bachelor of Engineering in Artificial Intelligence *Expected Dec 2027 GPA: 8.62/10.0*

- **Honors:** Certificate of Merit: Honorable Student of Trimester (Semesters 3, 4, 5, 6).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

PUBLICATIONS & MANUSCRIPTS

Co-Authors, **Gia-Hung Nguyen Le** (Second Author) *et al.*, "Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia."
Under Review at the Australasian Joint Conference on Artificial Intelligence (AJCAI 2026).

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, "Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring."
Under Review at the Australasian Joint Conference on Artificial Intelligence (AJCAI 2026).

Co-Authors, **Gia-Hung Nguyen Le** (Second Author) *et al.*, "Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution."
Under Review at the 33rd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2027).

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, "Proactive Air Quality Forecasting and Health Alert System for Melbourne."
Australasian Joint Conference on Artificial Intelligence (AJCAI 2025). Springer, Lecture Notes in Artificial Intelligence (LNAI). *Q2 (SJR 2024)*.
DOI: 10.1007/978-981-95-4969-6_34

RESEARCH & TECHNICAL EXPERIENCE

Health-Centred Air Quality Alert System (NZ & AU) Academic Research
Second Author · AJCAI 2026 Under Review *Jan 2026 – Present*

- Re-framed warning evaluation from concentration error to exposure coverage ($E_{s,d}$) across 10-year Canterbury ($n = 620,057$) and 4-year Melbourne ($n = 823,009$) hourly records.
- Proved regulatory daily-mean triggers overstate protection by 7.7–18.6 points, as 42% of exposure occurs on sub-threshold daily-mean days.
- Built an archived weather-forecast panel model reaching **MAE 3.19 μg/m^3** and R^2 **0.54** 1-day ahead (+26% over persistence, +12% over XGBoost), holding margin to 3 days.

Cyanobacterial Bloom Forecasting under Sparse Monitoring Academic Research
First Author & Corresponding Author · AJCAI 2026 Under Review *Jan 2026 – Present*

- Stress-tested multi-source time-series forecasting across 7 NZ lakes using a causal weekly benchmark ($n = 5,432$) and untouched 2023–2024 holdout ($n = 170$).
- Discovered horizon-dependent fusion dynamics: 7-day Chl-a history XGBoost reaches **weighted F1 0.791** (+0.103 over full fusion, $p = 0.016$), whereas 14-day full fusion reaches 0.703.
- Executed a staged incident-eligibility audit showing 22 apparent episode starts reduce to 3 true incidents; formulated an 8-gate provenance-aware audit protocol.
- Lead-authored manuscript and spearheaded experimental protocols and open-source data release.

Trustworthy Multimodal Anomaly Detection in Time Series Academic Research
Second Author · ACM SIGKDD 2027 Under Review *Jan 2026 – Present*

- Audited look-ahead text leakage across 9 real domains, proving naive text fusion creates an illusory 4.52% error reduction that reverses under embargo (leakage fraction 1.50).
- Formulated the **C-DLE** framework using **split-conformal calibration** and **Mondrian pools** for distribution-free false alarm control under missing modalities.
- Designed a cross-modal counterfactual sufficiency rule achieving **94.3% attribution accuracy** against ground truth anomalies, pruned via a 10% coreset.

Proactive Air Quality Forecasting System (SmokeNet) — Academic Research
First Author & Corresponding Author · AJCAI 2025 Published (LNAI Q2) — *Jun 2025 – Nov 2025*

- Designed and implemented **SmokeNet**, a Transformer-based architecture for $PM_{2.5}$ time-series forecasting.
- Achieved **MAE 0.7470** and R^2 **0.9545** (35.9% error reduction vs. XGBoost); outperformed baselines by **57.7%** during severe smoke backtests.
- Engineered end-to-end pipeline processing 4+ years of data with GPU cross-validation and integrated Explainable AI (SHAP, Integrated Gradients).
- First-authored full conference manuscript for AJCAI 2025 and released open-source reproducible evaluation suite.

MERR-GAT: Multimodal Emotion Recognition via Graph Attention — Academic Research
Second Author & Corresponding Author · Targeting Elsevier Information Fusion (Q1) — *Oct 2025 – Present*

- Developed **MERR-GAT**, a hybrid framework fusing **Wav2Vec 2.0** (audio) and **RoBERTa** (text) via Graph Attention Networks (GATv2) over dynamic utterance graphs.
- Implemented intrinsic XAI where learned GAT attention weights provide granular interpretability into conversational decision-making.
- Benchmarked on **IEMOCAP**, **RAVDESS**, and **MELD** with a dynamic classification head supporting 4, 5, or 6 emotion classes.

End-to-End Vietnamese Speech Recognition System — Academic Research
Co-Author · Speech Corpus Benchmark — *Sep 2025 – Present*

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.
- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.
- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets; benchmarked against **PhoWhisper**.

LEADERSHIP

SpeedyLabX - Student Research Group — FPT University
Founder & Student Lead — *2025 – Present*

- Founded a student research group of 10 undergraduates; led weekly technical workshops and reading groups.
- Coordinated submission workflow (milestones, reviews, camera-ready) for the group's first AJCAI-accepted paper.

CERTIFICATIONS

- **Natural Language Processing –** DeepLearning.AI
- **Gradient to Production: MLOps & Model Serving –** Coursera
- **IBM Full Stack Software Developer –** IBM
- **AI Engineer Professional –** Packt
- **Data Science Fundamentals & Containerization (Docker/K8s) –** IBM

SKILLS

- **Languages:** Python, SQL (PostgreSQL).
- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers, PyTorch Geometric (PyG / GATv2), TensorFlow/Keras, Scikit-learn, OpenCV.
- **Explainable AI (XAI) & Evaluation:** SHAP, Integrated Gradients, Chronological CV, CTC Loss, Weights & Biases (W&B).
- **Data & MLOps/Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Linux, Streamlit.
- **Technical writing:** Blog posts and technical documentation (portfolio: hei-portfolio.vercel.app).